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/*
 * C Program to Implement a Queue using an Array
 */

#include <stdio.h>

#include <stdlib.h>

#define MAX 50

void insert();

void Delete();

void display();

int queue_array[MAX];

int rear = -1;

int front = -1;

int

main()

{

    int choice;

    while (1)

    {

        printf("1.Insert element to queue \n");

        printf("2.Delete element from queue \n");

        printf("3.Display all elements of queue \n");

        printf("4.Quit \n");

        printf("Enter your choice : ");

        scanf("%d", &choice);

```

```

switch (choice)
{
    case 1:
        insert();
        break;
    case 2:
        Delete();
        break;
    case 3:
        display();
        break;
    case 4:
        exit(1);
    default:
        printf("Wrong choice \n");
} /* End of switch */

} /* End of while */

return 0;

} /* End of main() */

void insert()
{
    int add_item;

    if (rear == MAX - 1)

```

```

printf("Queue Overflow \n");

else

{
    if (front == - 1)

        /*If queue is initially empty */

        front = 0;

        printf("Inset the element in queue : ");

        scanf("%d", &add_item);

        rear = rear + 1;

        queue_array[rear] = add_item;

    }

} /* End of insert() */


void Delete()

{

    if (front == - 1 || front > rear)

    {

        printf("Queue Underflow \n");

        return ;

    }

    else

    {

        printf("Element deleted from queue is : %d\n", queue_array[front]);

        front = front + 1;

    }

}

```

```
 } /* End of delete() */
```

```
void display()
```

```
{
```

```
    int i;
```

```
    if (front == - 1)
```

```
        printf("Queue is empty \n");
```

```
    else
```

```
    {
```

```
        printf("Queue is : \n");
```

```
        for (i = front; i <= rear; i++)
```

```
            printf("%d ", queue_array[i]);
```

```
        printf("\n");
```

```
    }
```

```
 } /* End of display() */
```